Mobina Pournemat

PH.D. CANDIDATE IN COMPUTER SCIENCE · UNIVERSITY OF MARYLAND

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RESEARCH INTERESTS

Large language models and reasoning — with emphasis on **efficient inference and reasoning**, **reliable and faithful model behavior**, and **rigorous evaluation and benchmarking**. Additional experience in multimodal generative models and computer vision.

EDUCATION

University of Maryland, College Park — College Park, MD, USA | Aug. 2024 – Present
Ph.D. in Computer Science — Advised by Prof. Soheil Feizi | GPA: 4.0 / 4.0

Sharif University of Technology — Tehran, Iran | Sep. 2018 – Feb. 2023
B.Sc. in Computer Engineering

PUBLICATIONS

[1] **Reasoning Under Uncertainty: Exploring Probabilistic Reasoning Capabilities of LLMs**
M. Pournemat, K. Rezaei, G. Sriramanan, A. Zarei, H. Eghbalzadeh, Y. Wang, J. Fu, F. Peng, Z. Wang, X. Feng, N. Noorshams, S. Feizi
EMNLP 2026 (Main Conference) · arXiv:2509.10739

[2] **SliderEdit: Continuous Image Editing with Fine-Grained Instruction Control**
A. Zarei, S. Basu, **M. Pournemat**, S. Nag, R. Rossi, S. Feizi
CVPR 2026 (Oral) · arXiv:2511.09715 · project page

[3] **Same Question, Different Answers: Evaluating LLM Reliability Beyond Accuracy**
K. Faghih, Y. Cheng, S. Saha, **M. Pournemat**, A. Gerami, S. Feizi
Under review · arXiv:2607.22554

RESEARCH EXPERIENCE

University of Maryland, College Park — College Park, MD, USA | Aug. 2024 – Present
Graduate Research Assistant — Advisor: Prof. Soheil Feizi

- **Efficient Reasoning through Cross-Model Latent Knowledge Transfer** — IN PROGRESS
  Reducing the inference cost of long chain-of-thought reasoning by transferring knowledge between models in latent space rather than through generated tokens, so that a smaller model inherits a stronger model's reasoning behavior at a fraction of the decoding budget.
- **Probabilistic Reasoning Capabilities of LLMs** — EMNLP 2026 MAIN
  Built a comprehensive benchmark and evaluation suite measuring how well frontier LLMs reason under uncertainty; ran large-scale experiments across open- and closed-weight models and characterized systematic failure modes, calibration gaps, and the effect of prompting and inference-time compute.
- **Reliability of LLMs Beyond Accuracy**
  Co-developed an evaluation framework quantifying how consistently LLMs answer semantically identical questions, exposing response variance that aggregate accuracy metrics conceal.
- **Fine-Grained Controllability in Image Editing** — CVPR 2026 ORAL
  Contributed to SliderEdit, which turns discrete instruction-based image editing into a continuous, per-instruction control interface for text-to-image diffusion models.

Institute of Science and Technology Austria (ISTA) — Klosterneuburg, Austria | Nov. 2023 – Jul. 2024
Machine Learning Intern — Supervisor: Prof. Johann Danzl

- **3D Semantic Segmentation of Brain Microscopy Images**
  Built an efficient pipeline that adapts the Segment Anything Model to volumetric microscopy data, replacing a manual annotation workflow with automated segmentation for neuroscience researchers.

Sharif University of Technology — Tehran, Iran Jul. 2022 – Oct. 2023

NLP Research Assistant — Advisor: Dr. Ehsaneddin Asgari

- **Persian Syntactic and Semantic Error Correction**
  Developed an auto-correction system for Persian text combining a BERT-based language model with rule-based modules for morphology and spacing.

Sharif University of Technology — Tehran, Iran Sep. 2021 – Jul. 2022

Computer Vision Research Assistant — Advisor: Dr. Mohammad Hossein Rohban

- **Anomaly Detection and Semantic Segmentation**
  Studied autoencoder-based anomaly detection and segmentation methods, and benchmarked reconstruction- and representation-based approaches on standard vision datasets.

INDUSTRY EXPERIENCE

Behin Bazaar Marketing Agency — Tehran, Iran Apr. 2023 – Oct. 2023

Data Analyst, Business Intelligence

- Designed and maintained the analytics database, built ETL and preprocessing pipelines, and delivered dashboards and recurring reports (SQL, Power BI) that informed campaign decisions.

AIMedic — Tehran, Iran Jul. 2021 – Sep. 2021

Machine Learning for Healthcare Intern

- Developed classification and semantic segmentation models for medical imaging data, covering data cleaning, model training, and evaluation.

SELECTED PROJECTS

- **Quantifying Attributional Bias in LLM Toxicity Detection** (Nov. 2025)
  Measured how toxicity scores for identical comments shift when attributed to different author identities, isolating a bias that standard toxicity benchmarks do not surface.
- **Benchmarking Multi-Stage Vulnerability Reasoning in LLMs** (Oct. 2025)
  Evaluated how explicit reasoning traces and post-hoc verification affect LLM accuracy on software vulnerability detection.
- **Explainable Detection of Online Sexism** (May 2022)
  SemEval-2023 Task 10: hierarchical classification and explanation of sexist content in social media text.
- **Recommender System Design** (May 2021)
  Built and compared content-based, collaborative-filtering, and hybrid recommenders, with offline evaluation of ranking quality across user cohorts.

TECHNICAL SKILLS

Programming: Python, Java, C, SQL, JavaScript, PHP

ML & LLMs: PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn, NumPy, Pandas; GPU training and inference, fine-tuning, in-context learning and prompting, benchmark design and evaluation

Data & Tools: MySQL, PostgreSQL, SQL Server, Power BI, Docker, Git, Linux, LaTeX

Web: HTML/CSS, jQuery, Bootstrap

TEACHING EXPERIENCE

Teaching assistant at the University of Maryland and Sharif University of Technology: **Introduction to Artificial Intelligence** (Fall 2024), **Data Structures and Algorithms** (Fall 2022), **Natural Language Processing** (Spring 2022), **Computer Simulation** (Fall 2021), **Probability and Statistics** (Spring 2021).

SELECTED COURSEWORK

Graduate (UMD): Natural Language Processing, Long-Context Language Models, Multimodal Foundation Models, LLM Security and Privacy — all A (4.0). **Undergraduate:** Artificial Intelligence (20/20), Data Structures and Algorithms (19.6/20), Machine Learning (Stanford CS229, audited).

LANGUAGES

Persian — native · English — proficient · Spanish — elementary · Turkish — elementary